Zero-Short Prompting:-

You given the AI an instruction without providing of the expected output.

Ex

Explain python dictionaries to a beginner using simple daugvage and one example

Mental model: "Here is the task.
Figure out how to do it"

Four Short prompting:-

You given the AI Examples of the pattern you want it to follow before presenting the new task.

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Example 1: Input : Happy
output : Positive

Example 2: Input : Terrible
Output : Negative

Now classify : Excellent

Mental model: "watch these examples, learn the pattern, and Apply it"

Chain-of-thought (CoT) prompting:

you ask the AI to work through a problem systematically rather than jump straight to a conclusion.

Example

A notebook costs ₹120. It have a 25% discount. Calculate the final Price. Show the Calculation and the mental model:

"Break the problem into useful steps before giving the result"

Distinction:

Zero-shot and Few-short describe whether Examples are provided. Chain-of-thought (COT) describes a reasoning-oriented approach, So it can be combined with either one.

You can ask for a concise Explanation of the method without requiring the model to reveal private internal reasoning.

3. Why Prompting does this matter?

Consider the ambiguous instruction:

"Make me a plan to learn AI."

What's missing?

- What kind of AI?
- What is You current skill level?
- How much time do you have?
- What is your Goal?
- What should the final plan look like?

A prompt can be grammatically correct and still be vague to produce a usful answer.

The key lesson is this:

Better promts reduce unnecessary guessing by giving the AI the Information, Examples and Constraints it needs.

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The five parts of a Strong "Zero-Short promt"

- Role: Who should the AI act as?
- Task: what should it do?
- Context: What does the AI need to About the learner?
- Output format: How should the responce be organized?
- Soccer criteria: How will we know the task is doen well?

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Zero Shot Prompting

Zero short prompting means giving an AI instruction to complect a task without providing input,- output Examples of the Desired behavior.

Example

WEAK PROMPT

Teach me python dictionaries.

STRONG PROMPT:

Act as my python mentor. I am a B.Tech Student and a complete beginner in python dictionaries. Teach me one concept at a time using Simple language, a code Example, and real-world use. Ask one practice question and check my understanding before moving forward.

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Important Note

Both prompts are zero-shot because neither includes Example conversations demonstrating the desired response.

The Five most important Parts

🧑 Role :

Tell the AI who it should act as

Ex Act as my python mentor

🎯 Task

Clearly state what you want the AI To Do.

Ex Teach me python Dictionaries

🧠 Context

Explain your current knowledge and needs

Ex I am complete beginner and need Simple Explanations

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:= Output format

Specify how the answer should be Organized.

Ex Explaination → code Example → → real-world use → practice Question

✓ Success Criteria:

Define what the leaner must demonstrate before moving forward.

Ex Explain the concept without help and answer a new question independently.

What You Should avoid.

⊗ Avoid vague instruction

Don't write only "teach me well (or) "Explain clearly". Specify what a good Explation includes.

⊗ Avoid combining everything into one sentence

Separate the role, task, output format, and rules into district instruction.

⊛ Avoid revealing unclear mastery conditions

"Move forward when I understand" is weaker than defining what you must explain and slove independently

⊛ Avoid confusing zero-short and few-short

Zero-shout use instructions without demonstrations. Few-Shot includes Examples of input and their Expected outputs

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Few-short Prompting

1. What is few-short prompting

Few-short prompting technique in which we provide an AI with a few input-output Examples so it can understand the pattern and follow it for a new task.

* Don't just tell the AI what to do; Show it Examples of what a good answer looks like.

Example

Zero-short:

Tell me whether 8 is even or odd

Few-short:

Example :- is 2 even or odd →

The number is even bec because 2 % 2 == 0.

Example2:-

is 3 even or odd →

The number is odd because

3 % 2 == 1

The Examples demonstrate the expected response patten.

Structure of a Few-Shot Prompt:

1. Instruction: Explain what the AI Should Do
2. Example1 : Show an input and its Excepted Output
3. Example2 : Show another input and output
4. New input : Give the AI a new task to complet

General patten :

Instruction + Examples + New Input ↓

AI Response

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My Practice Examples → If num is even (or) odd

Example 1 :-

Input : is 2 is even or odd

output : Yes, 2 is even number

Because 2 % 2 = 0.

Example 2:

Input : 3 is even or odd

output : 3 is odd number

Because 3 % 2 = 1

* When we given (or) ask to ai 8 is even or odd the reply is

8 is even number
Because 8 % 2 == 0.

This Example demostrate the pattern:

Question → answer → Support calculation

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When should use Few-Short

Few should prompting is use fealty when I want the AI to follow a particulars.

- Response style or tone
- output format
- Classification pattern
- Text transformation patterns
- Tutoring or feedback Process

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Important Rules

- Use examples that are relevant to the task
- Make sure the examples are correct
- Keep the examples to demonstrate the behavior clearly, with use different Example.
- Keep the example consistent with the desired output pattern
- Remember the examples in a prompt provide context for the current task, they do not permanently train the AI model

- Clearly Separate instructions, Examples, and new input.

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Common Mistakes to Avoid

1. Providing instructions without actual input-output demonstrations When few-short Example are required
2. Use incorrect or inconsistet
3. Example
3. Giving only a final label when the Desired should include an Explanation.
4. Using Examples unrelated to the task
5. Confussing few-short prompting with Zero-Short prompting

Final Takeaway :-

Few-Short prompting teaches the AI a desired response pattern throught Examples included in the prompting.

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CoT → Chain-of-Thought Prompting

CoT- prompting technique that encourages an AI model to solve complex problem using logical, step by step approaches.

In Simple words:-

Instead of asking AI to give only the final answer, we guide it to break a problem into smaller steps and explain the solution.

Why CoT is used -

- Improves structure problem-solving
- Helps handle complex tasks
- Encourages logical explanations
- makes calculation easier to understand
- Helps analyse multiple conditions

Example

Normal prompting

I have ₹500. I spend around ₹150 to Buy a Book, How much money remaining, OUTPUT: 350

COT prompting

I have ₹500. I spend around ₹150 to By a Book

Break a problem into small steps. show the calculation and provide the final answer.

output

you have = 500

you spent on Book = 150

Remaining Money = ₹500 - 150

Final answer = 350.

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CoT Prompting structure

Role :- Defines the role or identity the AI should act as :

Ex Act as an world no.1 programer in python

Task: Defines what you want AI to do

Ex Explain how list works.

Instructions :- Provides specific direction that AI must follows while completing the task.

Ex

1. Explain in simple language
2. Give one Example
3. Explain the output

Output

Provide a clear explaination and final answer.

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Common Mistakes to Avoid

1. Using CoT for every simple question
2. Giving confusing or contradictory instruction
3. Asking for unnecessary lengthy explanation
4. Assuming step-by-step explanation are always correct.
5. Forgetting to ask the model to verify its final answer.

One important professional note :

Modern prompt engineering often uses structure reasoning instructions, intermediate checks, and concise explanations.

You do not need to expose a model's private internal chain of thought to get reliable results.

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Structured Reasoning Rompting

Given AI a step-by-step process to follow so it can produce a more organized and useful answer.

Example

Normal prompting

Fix my phy thon code.

Structured Prompting

Act as a phython debugging tutor.

Follow these steps:

1. Identify the error.
2. Explain why it occurs.
3. Suggest a correction
4. Verify the Corrected Code.
5. Explain what I should learn

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Important Components :-

1. **Role** : Tell AI who it should act as.
TASK : Explain what you want AI to do
STEPS : Divide the task into smaller actions
Explanation : specify how the answer should be Explained
verification : Ask AI to check the result
learning check : Ask a Question to test understanding when appropriate.

What to Avoid.

- Giving vague instruction
- mixing many tasks without an order
- Forgetting to specify the Expected output
- skipping verification when correctness matters
- Given Unnecessary steps:

Important — Structured reasoning Prompting asks AI to following an organized process and provide Usefull Explation.